Lunar Lander Fixed-Point Calculations

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1 Calculations

1.1 Initial Values

$$\Delta t = 1 \text{ s},$$

$$S = 1000,$$

$$h_0 = 15000S,$$

$$h_0 = 15000(1000),$$

$$h_0 = 15000000 \text{ mm},$$

$$v_0 = -50S,$$

$$v_0 = -50(1000),$$

$$v_0 = -50000 \text{ mm/s},$$

$$g = -1.62S,$$

$$g = -1.62(1000),$$

$$g = -1620 \text{ mm/s}^2,$$

$$T = 4.00S,$$

$$T = 4.00(1000),$$

$$T = 4000 \text{ mm/s}^2,$$

$$f_0 = 120 \text{ kg},$$

$$b = 4 \text{ kg/s},$$

$$v_{\text{limit}} = -20S,$$

$$v_{\text{limit}} = -20(1000),$$

$$v_{\text{limit}} = -20000 \text{ mm/s},$$

$$v_{\text{safe}} = -5S,$$

$$v_{\text{safe}} = -5(1000),$$

$$v_{\text{safe}} = -5000 \text{ mm/s}.$$

1.2 Engine State

$$E_n = \begin{cases} 1, & f_n > 0 \wedge v_n < -20000, \\ 0, & f_n \leq 0 \vee v_n \geq -20000. \end{cases}$$

1.3 Update Equations

$$\begin{aligned}
a_n &= g + E_n T, \\
f_{n+1} &= \max\left(0, f_n - E_n b\right), \\
v_{n+1} &= v_n + a_n \Delta t, \\
h_{n+1} &= \max\left(0, h_n + v_{n+1}\Delta t\right), \\
t_{n+1} &= t_n + \Delta t.
\end{aligned}$$

1.4 First Iteration

$$\begin{aligned}
f_0 &> 0, \\
120 &> 0, \\
v_0 &< v_{\text{limit}}, \\
-50000 &< -20000, \\
E_0 &= 1.
\end{aligned}$$

$$\begin{aligned}
a_0 &= g + E_0 T, \\
&= -1620 + (1)(4000), \\
&= -1620 + 4000, \\
&= 2380 \text{ mm/s}^2.
\end{aligned}$$

$$\begin{aligned}
f_1 &= \max\left(0, f_0 - E_0 b\right), \\
&= \max\left(0, 120 - (1)(4)\right), \\
&= \max(0, 116), \\
&= 116 \text{ kg}.
\end{aligned}$$

$$\begin{aligned}
v_1 &= v_0 + a_0 \Delta t, \\
&= -50000 + (2380)(1), \\
&= -50000 + 2380, \\
&= -47620 \text{ mm/s}.
\end{aligned}$$

$$\begin{aligned}
h_1 &= \max\left(0, h_0 + v_1 \Delta t\right), \\
&= \max\left(0, 15000000 + (-47620)(1)\right), \\
&= \max(0, 14952380), \\
&= 14952380 \text{ mm}.
\end{aligned}$$

$$
\begin{aligned}
t_1 &= t_0 + \Delta t, \\
&= 0 + 1, \\
&= 1 \text{ s}.
\end{aligned}
$$

1.5 Second Iteration

$$
\begin{aligned}
f_1 &> 0, \\
116 &> 0,
\end{aligned}
$$

$$
\begin{aligned}
v_1 &< v_{\text{limit}}, \\
-47620 &< -20000,
\end{aligned}
$$

$$
E_1 = 1.
$$

$$
\begin{aligned}
a_1 &= g + E_1 T, \\
&= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2.
\end{aligned}
$$

$$
\begin{aligned}
f_2 &= \max\left(0, f_1 - E_1 b\right), \\
&= \max\left(0, 116 - (1)(4)\right), \\
&= 112 \text{ kg}.
\end{aligned}
$$

$$
\begin{aligned}
v_2 &= v_1 + a_1 \Delta t, \\
&= -47620 + (2380)(1), \\
&= -45240 \text{ mm/s}.
\end{aligned}
$$

$$
\begin{aligned}
h_2 &= \max\left(0, h_1 + v_2 \Delta t\right), \\
&= \max\left(0, 14952380 + (-45240)(1)\right), \\
&= 14907140 \text{ mm}.
\end{aligned}
$$

$$
\begin{aligned}
t_2 &= t_1 + \Delta t, \\
&= 1 + 1, \\
&= 2 \text{ s}.
\end{aligned}
$$

1.6 Third Iteration

$$\begin{aligned}
f_2 &> 0, \\
112 &> 0, \\
v_2 &< v_{\text{limit}}, \\
-45240 &< -20000, \\
E_2 &= 1. \\
a_2 &= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2. \\
f_3 &= \max\left(0, 112 - (1)(4)\right), \\
&= 108 \text{ kg}. \\
v_3 &= -45240 + (2380)(1), \\
&= -42860 \text{ mm/s}. \\
h_3 &= \max\left(0, 14907140 + (-42860)(1)\right), \\
&= 14864280 \text{ mm}. \\
t_3 &= 2 + 1, \\
&= 3 \text{ s}.
\end{aligned}$$

1.7 Fourth Iteration

$$\begin{aligned}
E_3 &= 1, \\
a_3 &= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2, \\
f_4 &= 108 - (1)(4), \\
&= 104 \text{ kg}, \\
v_4 &= -42860 + 2380, \\
&= -40480 \text{ mm/s}, \\
h_4 &= 14864280 - 40480, \\
&= 14823800 \text{ mm}, \\
t_4 &= 3 + 1, \\
&= 4 \text{ s}.
\end{aligned}$$

1.8 Fifth Iteration

$$
\begin{aligned}
E_4 &= 1, \\
a_4 &= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2, \\
f_5 &= 104 - (1)(4), \\
&= 100 \text{ kg}, \\
v_5 &= -40480 + 2380, \\
&= -38100 \text{ mm/s}, \\
h_5 &= 14823800 - 38100, \\
&= 14785700 \text{ mm}, \\
t_5 &= 4 + 1, \\
&= 5 \text{ s}.
\end{aligned}
$$

1.9 Sixth Iteration

$$
\begin{aligned}
E_5 &= 1, \\
a_5 &= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2, \\
f_6 &= 100 - (1)(4), \\
&= 96 \text{ kg}, \\
v_6 &= -38100 + 2380, \\
&= -35720 \text{ mm/s}, \\
h_6 &= 14785700 - 35720, \\
&= 14749980 \text{ mm}, \\
t_6 &= 5 + 1, \\
&= 6 \text{ s}.
\end{aligned}
$$

1.10 Seventh Iteration

$$
\begin{aligned}
E_6 &= 1, \\
a_6 &= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2, \\
f_7 &= 96 - (1)(4), \\
&= 92 \text{ kg}, \\
v_7 &= -35720 + 2380, \\
&= -33340 \text{ mm/s}, \\
h_7 &= 14749980 - 33340, \\
&= 14716640 \text{ mm}, \\
t_7 &= 6 + 1, \\
&= 7 \text{ s}.
\end{aligned}
$$

1.11 Eighth Iteration

$$
\begin{aligned}
E_7 &= 1, \\
a_7 &= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2, \\
f_8 &= 92 - (1)(4), \\
&= 88 \text{ kg}, \\
v_8 &= -33340 + 2380, \\
&= -30960 \text{ mm/s}, \\
h_8 &= 14716640 - 30960, \\
&= 14685680 \text{ mm}, \\
t_8 &= 7 + 1, \\
&= 8 \text{ s}.
\end{aligned}
$$

1.12 Ninth Iteration

$$
\begin{aligned}
E_8 &= 1, \\
a_8 &= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2, \\
f_9 &= 88 - (1)(4), \\
&= 84 \text{ kg}, \\
v_9 &= -30960 + 2380, \\
&= -28580 \text{ mm/s}, \\
h_9 &= 14685680 - 28580, \\
&= 14657100 \text{ mm}, \\
t_9 &= 8 + 1, \\
&= 9 \text{ s}.
\end{aligned}
$$

1.13 Tenth Iteration

$$
\begin{aligned}
E_9 &= 1, \\
a_9 &= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2, \\
f_{10} &= 84 - (1)(4), \\
&= 80 \text{ kg}, \\
v_{10} &= -28580 + 2380, \\
&= -26200 \text{ mm/s}, \\
h_{10} &= 14657100 - 26200, \\
&= 14630900 \text{ mm}, \\
t_{10} &= 9 + 1, \\
&= 10 \text{ s}.
\end{aligned}
$$

1.14 Conversion After Ten Seconds

$$
\begin{aligned}
h_{10} &= \frac{14630900}{1000}, \\
&= 14630.9 \text{ m}.
\end{aligned}
$$

$$
\begin{aligned}
v_{10} &= \frac{-26200}{1000}, \\
&= -26.2 \text{ m/s}.
\end{aligned}
$$

$$f_{10} = 80 \text{ kg},$$

$$t_{10} = 10 \text{ s}.$$

1.15 Engine-Off Acceleration

$$
\begin{aligned}
E_n &= 0, \\
a_n &= g + E_n T, \\
&= -1620 + (0)(4000), \\
&= -1620 \text{ mm/s}^2.
\end{aligned}
$$

1.16 Engine-On Acceleration

$$
\begin{aligned}
E_n &= 1, \\
a_n &= g + E_n T, \\
&= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2.
\end{aligned}
$$

1.17 Touchdown Conditions

$$h_n \leq 0$$

$$
\begin{aligned}
h_n &\leq 0, \\
v_n &\geq -5000
\end{aligned}
\quad \Longrightarrow \quad \text{Safe landing.}
$$

$$
\begin{aligned}
h_n &\leq 0, \\
v_n &< -5000
\end{aligned}
\quad \Longrightarrow \quad \text{Crash landing.}
$$

1.18 Successful Landing Scenario

$$
\begin{aligned}
\Delta t &= 1 \text{ s}, \\
S &= 1000, \\
h_0 &= 300S, \\
&= 300(1000), \\
&= 300000 \text{ mm}, \\
v_0 &= -20S, \\
&= -20(1000), \\
&= -20000 \text{ mm/s}, \\
f_0 &= 200 \text{ kg}, \\
g &= -1.62S, \\
&= -1620 \text{ mm/s}^2, \\
T &= 4.00S, \\
&= 4000 \text{ mm/s}^2, \\
b &= 4 \text{ kg/s}, \\
v_{\text{limit}} &= -5S, \\
&= -5000 \text{ mm/s}, \\
v_{\text{safe}} &= -5S, \\
&= -5000 \text{ mm/s}.
\end{aligned}
$$

1.19 Simulation Equations

$$
E_n = \begin{cases} 1, & f_n > 0 \land v_n < -5000, \\ 0, & f_n \leq 0 \lor v_n \geq -5000. \end{cases}
$$

$$
\begin{aligned}
a_n &= g + E_n T, \\
f_{n+1} &= \max\left(0, f_n - E_n b\right), \\
v_{n+1} &= v_n + a_n \Delta t, \\
h_{n+1} &= \max\left(0, h_n + v_{n+1} \Delta t\right), \\
t_{n+1} &= t_n + \Delta t.
\end{aligned}
$$

2 Selected Successful-Landing Data

Table 1: Selected states from the successful landing simulation

$t\,(\mathrm{s})$	E	$a\,(\mathrm{m/s^2})$	$f\,(\mathrm{kg})$	$v\,(\mathrm{m/s})$	$h\,(\mathrm{m})$
0	–	–	200	-20.00	300.00
1	1	2.38	196	-17.62	282.38
2	1	2.38	192	-15.24	267.14
3	1	2.38	188	-12.86	254.28
4	1	2.38	184	-10.48	243.80
5	1	2.38	180	-8.10	235.70
6	1	2.38	176	-5.72	229.98
7	1	2.38	172	-3.34	226.64
8	0	-1.62	172	-4.96	221.68
9	0	-1.62	172	-6.58	215.10
10	1	2.38	168	-4.20	210.90
15	0	-1.62	160	-4.30	189.60
20	0	-1.62	152	-4.40	167.80
25	0	-1.62	144	-4.50	145.50
30	0	-1.62	136	-4.60	122.70
35	0	-1.62	128	-4.70	99.40
40	0	-1.62	120	-4.80	75.60
45	0	-1.62	112	-4.90	51.30
50	0	-1.62	104	-5.00	26.50
51	0	-1.62	104	-6.62	19.88
52	1	2.38	100	-4.24	15.64
53	0	-1.62	100	-5.86	9.78
54	1	2.38	96	-3.48	6.30
55	0	-1.62	96	-5.10	1.20
56	1	2.38	92	-2.72	0.00

2.1 First Iteration

$$f_0 > 0,$$

$$200 > 0,$$

$$v_0 < v_{\text{limit}},$$

$$-20000 < -5000,$$

$$E_0 = 1.$$

$$
\begin{aligned}
a_0 &= g + E_0 T, \\
&= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2.
\end{aligned}
$$

$$
\begin{aligned}
f_1 &= \max\left(0, 200 - (1)(4)\right), \\
&= 196 \text{ kg}.
\end{aligned}
$$

$$
\begin{aligned}
v_1 &= v_0 + a_0 \Delta t, \\
&= -20000 + (2380)(1), \\
&= -17620 \text{ mm/s}, \\
&= -17.62 \text{ m/s}.
\end{aligned}
$$

$$
\begin{aligned}
h_1 &= h_0 + v_1 \Delta t, \\
&= 300000 + (-17620)(1), \\
&= 282380 \text{ mm}, \\
&= 282.38 \text{ m}.
\end{aligned}
$$

2.2 Engine Deactivation

$$
\begin{aligned}
f_7 &= 172 \text{ kg}, \\
v_7 &= -3340 \text{ mm/s}, \\
v_{\text{limit}} &= -5000 \text{ mm/s}.
\end{aligned}
$$

$$
\begin{aligned}
v_7 &\geq v_{\text{limit}}, \\
-3340 &\geq -5000, \\
E_7 &= 0.
\end{aligned}
$$

$$
\begin{aligned}
a_7 &= g + (0)T, \\
&= -1620 \text{ mm/s}^2.
\end{aligned}
$$

$$
\begin{aligned}
f_8 &= 172 - (0)(4), \\
&= 172 \text{ kg}.
\end{aligned}
$$

$$
\begin{aligned}
v_8 &= -3340 - 1620, \\
&= -4960 \text{ mm/s}, \\
&= -4.96 \text{ m/s}.
\end{aligned}
$$

$$
\begin{aligned}
h_8 &= 226640 - 4960, \\
&= 221680 \text{ mm}, \\
&= 221.68 \text{ m}.
\end{aligned}
$$

2.3 Engine Reactivation

$$
\begin{aligned}
f_9 &= 172 \text{ kg}, \\
v_9 &= -6580 \text{ mm/s}.
\end{aligned}
$$

$$
\begin{aligned}
f_9 &> 0, \\
v_9 &< v_{\text{limit}}, \\
-6580 &< -5000, \\
E_9 &= 1.
\end{aligned}
$$

$$
\begin{aligned}
a_9 &= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2.
\end{aligned}
$$

$$
\begin{aligned}
f_{10} &= 172 - (1)(4), \\
&= 168 \text{ kg}.
\end{aligned}
$$

$$
\begin{aligned}
v_{10} &= -6580 + 2380, \\
&= -4200 \text{ mm/s}, \\
&= -4.20 \text{ m/s}.
\end{aligned}
$$

$$
\begin{aligned}
h_{10} &= 215100 - 4200, \\
&= 210900 \text{ mm}, \\
&= 210.90 \text{ m}.
\end{aligned}
$$

2.4 State Before Touchdown

$$
\begin{aligned}
t_{55} &= 55 \text{ s}, \\
h_{55} &= 1200 \text{ mm}, \\
v_{55} &= -5100 \text{ mm/s}, \\
f_{55} &= 96 \text{ kg}.
\end{aligned}
$$

$$
\begin{aligned}
f_{55} &> 0, \\
v_{55} &< v_{\text{limit}}, \\
-5100 &< -5000, \\
E_{55} &= 1.
\end{aligned}
$$

2.5 Final Acceleration

$$
\begin{aligned}
a_{55} &= g + E_{55}T, \\
&= -1620 + (1)(4000), \\
&= 2380 \text{ mm/s}^2.
\end{aligned}
$$

2.6 Final Fuel Calculation

$$
\begin{aligned}
f_{56} &= \max\left(0, f_{55} - E_{55}b\right), \\
&= \max\left(0, 96 - (1)(4)\right), \\
&= 92 \text{ kg}.
\end{aligned}
$$

2.7 Final Velocity Calculation

$$
\begin{aligned}
v_{56} &= v_{55} + a_{55}\Delta t, \\
&= -5100 + (2380)(1), \\
&= -2720 \text{ mm/s}, \\
&= \frac{-2720}{1000}, \\
&= -2.72 \text{ m/s}.
\end{aligned}
$$

2.8 Final Altitude Calculation

$$
\begin{aligned}
h^*_{56} &= h_{55} + v_{56}\Delta t, \\
&= 1200 + (-2720)(1), \\
&= -1520 \text{ mm}.
\end{aligned}
$$

$$
\begin{aligned}
h_{56} &= \max\left(0, h^*_{56}\right), \\
&= \max(0, -1520), \\
&= 0 \text{ mm}.
\end{aligned}
$$

2.9 Landing Classification

$$
\begin{aligned}
h_{56} &= 0, \\
v_{56} &= -2720 \text{ mm/s}, \\
v_{\text{safe}} &= -5000 \text{ mm/s}. \\
-2720 &\geq -5000, \\
v_{56} &\geq v_{\text{safe}}, \\
\therefore \quad \text{Result} &= \text{Safe landing}.
\end{aligned}
$$

2.10 Final Mission State

$$
\begin{aligned}
t_{\text{final}} &= 56 \text{ s}, \\
h_{\text{final}} &= 0 \text{ m}, \\
v_{\text{final}} &= -2.72 \text{ m/s}, \\
f_{\text{final}} &= 92 \text{ kg}.
\end{aligned}
$$